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Uncertainty-Aware Bayesian Prompt Imputation and Recovery for Sparse Multi-Prompt LLM Evaluation

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Abstract

Large language model evaluation is usually reported as a single benchmark accuracy under a single prompt template. Despite recent work argue that prompt sensitivity exists and that prompt choice can substantially affect measured performance and model ranking. Multi-prompt evaluation replaces this unstable scalar by the empirical distribution of performance across multi-prompt templates. But exhaustive evaluations on every prompt-item becomes expensive. In this paper, we study the problem of estimating the full distribution of model performance over many prompt variants given only a subset of observed prompt-item outcomes. We introduce BayPIRE, a Bayesian Prompt Imputation and Recovery Evaluation framework based on additive item response models. BayPIRE borrows strength across prompts and items for missing cells and returns posterior predictive intervals for realised finite-benchmark prompt scores. We compare BayPIRE with naive prompt averaging and the official PromptEval Rasch estimator on MMLU benchmarks. In our experiment, BayPIRE gives the lowest prompt-score error, Wasserstein distributional error, hidden-cell log loss, and hidden-cell brier score at all budgets. Posterior predictive coverage is conservative but stable for nominal 95% prompt-score intervals. The results support Bayesian item response modelling as an effective uncertainty-aware recovery estimator on multi-prompt evaluation.

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1 Introduction

Large language models are commonly summarised, compared and ranked by a single benchmark score. A model is often reported as achieving a certain accuracy on multiple choice tasks, a solve rate on mathematical reasoning, or a leaderboard aggregate. These numbers are convenient but they can create misleading results. A score depends not only on the model and benchmark items, but also on the prompt template used for evaluation. Therefore, a small rephrase in the wording will vary the scores. This means that a single-prompt evaluation is only one sample from a much broader protocol.

Recent studies on prompt sensitivity shows that evaluating LLMs on benchmarks using a single prompt can be unreliable and produce misleading conclusions about model ability [2]. A more robust evaluation is therefore the distribution of scores over a multiple prompts pool. This distribution can describe average performance, lower-tail risk under bad prompt choices, and upper-tail performance under good prompt selection. But the difficulty is cost. With I prompts and J benchmark items, exhaustive multi-prompt evaluation requires $I \times J$ model calls for every model and subject.

PromptEval studies this sparse recovery problem using item-response models [1]. BayPIRE keeps the same finite multi-prompt recovery target, which are the realised prompt scores and their empirical distribution under a fixed evaluation procedure. The difference is that BayPIRE also quantifies uncertainty about the unobserved prompt-item outcomes. Its posterior predictive intervals should therefore be interpreted as uncertainty about hidden realised cells.

Our experiments align with the PromptEval MMLU setting. We use a fixed pool of 100 prompt templates, budgets $B \in \{200, 400, 800, 1600\}$, and two-way balanced masking as the main sampling design. We evaluate five released LLM result sets over ten MMLU subjects. Uniform random masking is reported as a robustness check in the appendix.

Contributions. This paper makes four contributions.

1. **Finite-target formulation.** We formalise sparse multi-prompt LLM evaluation as recovery of the realised empirical distribution of prompt scores F^* and its quantiles under a fixed evaluation protocol.
2. **Bayesian prompt-recovery estimator.** We introduce BayPIRE, a Bayesian additive item-response estimator that uses posterior mean probabilities for point recovery, and uses posterior predictive Bernoulli draws for uncertainty quantification.
3. **Uncertainty-aware evaluation.** We evaluate prompt-score error, Wasserstein distributional error, hidden-cell log loss, hidden-cell brier score, posterior predictive coverage, and interval width.
4. **Empirical validation.** We demonstrate the efficacy of BayPIRE over Avg and the official PromptEval Rasch estimator under same masks, budgets, and metrics on MMLU recovery experiments.

2 Literature Review

Prompt sensitivity and multi-prompt evaluation. Prompt sensitivity is now a central concern in LLM evaluation. Mizrahi et al. [2] argue that evaluation should not rely on a single prompt template because prompt variation can affect measured performance and relative model

rankings. Sclar et al. [3] further show that small, meaning-preserving formatting changes can lead to large model performance change. This motivates reporting summaries over a prompt distribution rather than a single prompt score.

Item response theory. Item response theory provides a natural statistical language for sparse prompt–item evaluation because it separates respondent ability from item difficulty. The Rasch model [8] introduced the canonical one-parameter logistic model for binary item responses. Cai et al. [9] provide a modern review of IRT models and estimation, while Van der Linden [10] gives a comprehensive handbook treatment. In LLM evaluation, prompts or systems can be treated as responding units and benchmark questions as items. Prior NLP [11] work has also used IRT to study benchmark examples and leaderboard reliability. BayPIRE is a bayesian extension of an IRT model.

Efficient multi-prompt estimation. PromptEval [1] is the closest method to BayPIRE. It defines a binary prompt–item correctness table, prompt scores, an empirical performance distribution, and a quantile function. It then uses item-response modelling and sparse sampling designs to recover these quantities from a limited evaluation budget. BayPIRE follows this finite distributional target and the two-way balanced design, but replaces pure point estimation with Bayesian posterior predictive recovery.

Uncertainty in LLM evaluation. Several recent papers suggest that LLM evaluations should be analysed as statistical experiments rather than deterministic leaderboard entries. Miller [5] emphasises uncertainty quantification for evaluation results. Bowyer et al. [6] argue that normal approximations can be unreliable for small benchmark samples. This perspective is particularly relevant for sparse multi-prompt evaluation, where many prompt–item outcomes are unobserved.

Research gap. Existing work motivates prompt-sensitive evaluation, efficient sparse recovery, and uncertainty-aware benchmark reporting. The remaining gap is to recover both the prompt-performance distribution and the uncertainty induced by unobserved prompt–item outcomes. BayPIRE addresses this gap with a Bayesian item-response recovery framework.

3 Background

This section introduces the finite recovery problem. We consider a fixed LLM evaluated on a benchmark with J items using a fixed pool of I prompt templates. Exhaustive multi-prompt evaluation would run every prompt on every item. Our objective is to recover the distribution of prompt-level scores from only a small subset of outcomes, while quantifying the uncertainty caused by the unobserved evaluations.

3.1 Finite Prompt–Item Evaluation

Let $\mathcal{I} = [I]$ be a finite prompt set and let $\mathcal{J} = [J]$ be a finite item set. After running each prompt–item pair through the LLM and grading the response using the benchmark grading system, we obtain a binary correctness outcome $Y_{ij}^* \in \{0, 1\}$ ¹, where $Y_{ij}^* = 1$ if prompt i yields

¹In stochastic evaluation settings, the outcome can instead be a value in $[0, 1]$; see Appendix A.

a correct answer on item j and $Y_{ij}^* = 0$ otherwise. The full exhaustive multi-prompt evaluation is represented by the realised binary correctness matrix

$$Y^* = (Y_{ij}^*)_{i \in \mathcal{I}, j \in \mathcal{J}} \in \{0, 1\}^{I \times J}.$$

Definition 3.1 (Prompt score). The realised score of prompt i is

$$S_i^* = \frac{1}{J} \sum_{j=1}^J Y_{ij}^*.$$

Definition 3.2 (Prompt-score distribution). The finite empirical distribution and quantile function of prompt performance are

$$F^*(x) = \frac{1}{I} \sum_{i=1}^I \mathbf{1}\{S_i^* \leq x\}, \quad Q^*(u) = \inf\{x \in \mathbb{R} : F^*(x) \geq u\}.$$

3.2 Sparse Observation

In exhaustive evaluation, running all $I \times J$ prompt–item pairs is often expensive. We therefore observe only

$$\mathcal{E} \subseteq \mathcal{I} \times \mathcal{J}, \quad |\mathcal{E}| = B,$$

and the observed outcomes are

$$Y_{\mathcal{E}} = \{Y_{ij}^* : (i, j) \in \mathcal{E}\}.$$

The sparsity level is $\rho = B/(IJ)$, so that smaller ρ corresponds to a sparser evaluation.

In our MMLU experiment, we use 100 prompts and only observe $B \in \{200, 400, 800, 1600\}$ prompt–item pairs in our experiments, so only a small part of the full evaluation is revealed.

3.3 Objective and Evaluation Criteria

Given only $Y_{\mathcal{E}}$, our objective is to estimate the realised prompt scores S_i^* , the empirical distribution F^* , the quantiles Q^* and quantify missing-data uncertainty.

We evaluate sparse recovery along four dimensions.

1. **Point recovery:** asks whether recovered prompt scores are close to realised prompt scores.
2. **Distributional recovery:** asks whether the recovered empirical distribution over prompts is close to F^* .
3. **Hidden-cell probabilistic recovery:** asks whether fitted probabilities are accurate for unobserved binary outcomes.
4. **Uncertainty quality:** asks whether BayPIRE’s posterior predictive intervals cover the realised prompt scores and whether their width decreases as the budget increases.

4 Methods

4.1 Sampling Design

For each recovery experiment, an observed set $\mathcal{E}$ is generated with budget B . The main design is two-way balanced sampling, following PromptEval [1]. It spreads observations across both prompts and items so that the budget is not concentrated in a small part of the table².

Algorithm 1: Two-way balanced sampling

Input: Prompt set $\mathcal{I}$, item set $\mathcal{J}$, budget B

Output: Observed set $\mathcal{E}$

Initialise $\mathcal{E} \leftarrow \emptyset$;

Initialise prompt counts $d_i \leftarrow 0$ and item counts $c_j \leftarrow 0$;

for $b = 0$ **to** $B - 1$ **do**

 Among prompts with the least d_i , randomly choose $\hat{i}$;

 Among items j with $(\hat{i}, j) \notin \mathcal{E}$, randomly choose $\hat{j}$ from those with the least c_j ;

 Update $\mathcal{E} \leftarrow \mathcal{E} \cup \{(\hat{i}, \hat{j})\}$;

 Update $d_{\hat{i}} \leftarrow d_{\hat{i}} + 1$, $c_{\hat{j}} \leftarrow c_{\hat{j}} + 1$;

return $\mathcal{E}$

4.2 Observed-Retention Reconstruction

All methods use observed-retention reconstruction rule. Observed entries are kept equal to their realised values, and unobserved entries are imputed. If $\hat{p}_{ij}$ is the fitted probability of correctness, then

$$\hat{Y}_{ij} = \begin{cases} Y_{ij}^*, & (i, j) \in \mathcal{E}, \\ \hat{p}_{ij}, & (i, j) \in \mathcal{E}^c. \end{cases}$$

The recovered prompt score is

$$\hat{S}_i = \frac{1}{J} \sum_{j=1}^J \hat{Y}_{ij}.$$

The recovered empirical distribution and quantile function are

$$\hat{F}(x) = \frac{1}{I} \sum_{i=1}^I \mathbf{1}\{\hat{S}_i \leq x\}, \quad \hat{Q}(u) = \inf\{x \in \mathbb{R} : \hat{F}(x) \geq u\}.$$

BayPIRE uses fitted-probability reconstruction for point recovery, while uncertainty quantification uses posterior predictive Bernoulli reconstructions that also retain observed entries.³

4.3 Item-Response Model

The model-based estimators use an additive item-response structure. Let p_{ij} denote the probability that prompt i answers item j correctly. The Rasch form is

$$p_{ij}(\theta) = \text{logit}^{-1}(\alpha + a_i - b_j), \quad \theta = (\alpha, a_1, \dots, a_I, b_1, \dots, b_J). \quad (1)$$

²Uniform random sampling is used as a robustness check; see Appendix B.

³For detailed posterior predictive Bernoulli reconstruction rule; see Section 4.6.

where α is a global intercept, a_i is a prompt effect, and b_j is an item difficulty effect. Larger a_i increases the probability of correctness, while larger b_j decreases it. Therefore, it can borrow information across observed prompt–item outcomes.

4.4 Baseline Estimators

We compare BayPIRE with two baselines. Both baselines use the same observed set $\mathcal{E}$ and the same observed-retention reconstruction rule.

Avg. The simplest baseline estimates each prompt score by the average of the observed outcomes for that prompt. Let $\mathcal{E}_i = \{j \in \mathcal{J} : (i, j) \in \mathcal{E}\}$. If $|\mathcal{E}_i| > 0$, then missing cells in row i use

$$\hat{p}_{ij}^{\text{Avg}} = \frac{1}{|\mathcal{E}_i|} \sum_{k \in \mathcal{E}_i} Y_{ik}^*.$$

If $|\mathcal{E}_i| = 0$, the global observed mean is used. Avg is a minimal sanity-check baseline because it ignores item difficulty.

PE-Rasch. PE-Rasch is the Rasch-based sparse-evaluation estimator introduced by PromptEval [1]. It models correctness using the additive item-response model in Equation (1), where

$$\theta_{\text{PE}} = (\alpha, a_1, \dots, a_I, b_1, \dots, b_J), \quad b_{j_0} = 0.$$

The parameters are fitted by maximum likelihood on the observed outcomes

$$\hat{\theta}_{\text{PE}} = \arg \max_{\theta} \sum_{(i,j) \in \mathcal{E}} [Y_{ij}^* \log p_{ij}(\theta) + (1 - Y_{ij}^*) \log \{1 - p_{ij}(\theta)\}].$$

The fitted probabilities

$$\hat{p}_{ij}^{\text{PE}} = p_{ij}(\hat{\theta}_{\text{PE}})$$

are inserted into the observed-retention reconstruction rule to produce $\hat{S}_i$, $\hat{F}$, and $\hat{Q}$.⁴ Therefore, PE-Rasch is a point-estimation method. It conditions on a single fitted parameter vector and does not propagate posterior parameter uncertainty.

4.5 BayPIRE Model

BayPIRE places a hierarchical Bayesian prior on the additive Rasch model. For each observed prompt–item pair,

$$Y_{ij}^* \mid \alpha, a_i, b_j \sim \text{Bernoulli}\{p_{ij}(\theta)\}, \quad (i, j) \in \mathcal{E},$$

where $p_{ij}(\theta)$ is defined in Equation (1). The intercept has prior

$$\alpha \sim N(0, \sigma_\alpha^2), \quad \sigma_\alpha = 5.$$

For the prompt effects, BayPIRE introduces raw effects $a_i^{\text{raw}} \stackrel{\text{iid}}{\sim} N(0, 1)$ and defines

$$a_i = \sigma_a \left(a_i^{\text{raw}} - \frac{1}{I} \sum_{k=1}^I a_k^{\text{raw}} \right).$$

⁴For further details on the underlying PromptEval framework; see PromptEval [1].

Similarly, $b_j^{\text{raw}} \stackrel{\text{iid}}{\sim} N(0, 1)$ and

$$b_j = \sigma_b \left(b_j^{\text{raw}} - \frac{1}{J} \sum_{\ell=1}^J b_\ell^{\text{raw}} \right).$$

This construction enforces $\sum_i a_i = 0$ and $\sum_j b_j = 0$, resolving the additive non-identifiability. The scale parameters are assigned half-normal priors

$$\sigma_a \sim \text{HalfNormal}(\tau), \quad \sigma_b \sim \text{HalfNormal}(\tau), \quad \tau = 1$$

Posterior inference is performed using the No-U-Turn Sampler [13], an adaptive Hamiltonian Monte Carlo method [12]. We use 1000 tuning draws, 1000 retained posterior draws, four chains, and target acceptance probability 0.97. During tuning, NUTS adapts the step size and mass matrix automatically.

4.6 Posterior Point and Predictive Recovery

Let $\theta^{(1)}, \dots, \theta^{(M)}$ denote posterior draws. For point recovery, BayPIRE uses posterior mean correctness probabilities

$$\hat{p}_{ij} = \mathbb{E}[p_{ij} \mid Y_{\mathcal{E}}] \approx \frac{1}{M} \sum_{m=1}^M p_{ij}(\theta^{(m)}),$$

and then applies observed-retention reconstruction.

For uncertainty, BayPIRE uses posterior predictive Bernoulli reconstruction. For each posterior draw m , observed cells are retained exactly and missing cells are sampled:

$$\tilde{Y}_{ij}^{(m)} = \begin{cases} Y_{ij}^*, & (i, j) \in \mathcal{E}, \\ Z_{ij}^{(m)}, & (i, j) \in \mathcal{E}^c, \end{cases} \quad Z_{ij}^{(m)} \sim \text{Bernoulli}\{p_{ij}(\theta^{(m)})\}.$$

The corresponding posterior predictive prompt score is

$$S_i^{(m)} = \frac{1}{J} \sum_{j=1}^J \tilde{Y}_{ij}^{(m)}.$$

The empirical 2.5% and 97.5% quantiles of $S_i^{(1)}, \dots, S_i^{(M)}$ define a nominal 95% posterior predictive interval for prompt i ,

$$[L_i^{0.95}, U_i^{0.95}] = [Q_{0.025}(S_i^{(1:M)}), Q_{0.975}(S_i^{(1:M)})].$$

4.7 Evaluation Metrics

Let S_i^* be the realised prompt score and $\hat{S}_i$ the recovered prompt score.

Definition 4.1 (Prompt-level point recovery). Mean absolute error and root mean squared error are

$$\text{MAE} = \frac{1}{I} \sum_{i=1}^I |\hat{S}_i - S_i^*|, \quad \text{RMSE} = \left\{ \frac{1}{I} \sum_{i=1}^I (\hat{S}_i - S_i^*)^2 \right\}^{1/2}.$$

Definition 4.2 (Distributional recovery). Distributional recovery is evaluated using

$$W_1(\hat{F}, F^*) = \int_0^1 |\hat{Q}(u) - Q^*(u)| du,$$

where $\hat{Q}$ and Q^* are the corresponding quantile functions.

Definition 4.3 (Hidden-cell probability prediction). On $\mathcal{E}^c$, the hidden-cell log loss is

$$\text{LogLoss} = -\frac{1}{|\mathcal{E}^c|} \sum_{(i,j) \in \mathcal{E}^c} [Y_{ij}^* \log \hat{p}_{ij} + (1 - Y_{ij}^*) \log(1 - \hat{p}_{ij})],$$

and the hidden-cell brier score is

$$\text{Brier} = \frac{1}{|\mathcal{E}^c|} \sum_{(i,j) \in \mathcal{E}^c} (Y_{ij}^* - \hat{p}_{ij})^2.$$

Definition 4.4 (Posterior predictive uncertainty). For BayPIRE, let $[L_i^{0.95}, U_i^{0.95}]$ be the nominal 95% posterior predictive interval for prompt i . Predictive coverage and mean interval width are

$$\text{Coverage}_{0.95} = \frac{1}{I} \sum_{i=1}^I \mathbf{1}\{S_i^* \in [L_i^{0.95}, U_i^{0.95}]\}, \quad \text{Width}_{0.95} = \frac{1}{I} \sum_{i=1}^I (U_i^{0.95} - L_i^{0.95}).$$

5 Theoretical Properties

This section records basic theoretical properties of the BayPIRE reconstruction. The results are conditional on the fixed prompt set, item set, and observed design $\mathcal{E}$.

Proposition 5.1 (Bayes optimality under squared loss). *Among all estimators $T_i = T_i(Y_{\mathcal{E}})$ of S_i^* , the posterior mean*

$$\hat{S}_i^{\text{BayPIRE}} = \mathbb{E}[S_i^* | Y_{\mathcal{E}}]$$

minimises posterior expected squared error:

$$\hat{S}_i^{\text{BayPIRE}} = \arg \min_{T_i} \mathbb{E}[(T_i - S_i^*)^2 | Y_{\mathcal{E}}].$$

Proof. For any estimator T_i ,

$$\mathbb{E}[(T_i - S_i^*)^2 | Y_{\mathcal{E}}] = (T_i - \mathbb{E}[S_i^* | Y_{\mathcal{E}}])^2 + \text{Var}(S_i^* | Y_{\mathcal{E}}).$$

Therefore, the posterior expected squared error is minimised by

$$T_i = \mathbb{E}[S_i^* | Y_{\mathcal{E}}].$$

□

Theorem 5.1 (Posterior predictive variance decomposition). *Let*

$$\mathcal{E}_i^c = \{j \in \mathcal{J} : (i, j) \notin \mathcal{E}\}.$$

Conditional on the correctness probabilities p_{ij} , suppose the unobserved outcomes are independent Bernoulli random variables. Then

$$\text{Var}(S_i^* | Y_{\mathcal{E}}) = \frac{1}{J^2} \left[\mathbb{E} \left\{ \sum_{j \in \mathcal{E}_i^c} p_{ij}(1 - p_{ij}) \mid Y_{\mathcal{E}} \right\} + \text{Var} \left(\sum_{j \in \mathcal{E}_i^c} p_{ij} \mid Y_{\mathcal{E}} \right) \right].$$

Proof. Observed entries are fixed after conditioning on $Y_{\mathcal{E}}$, so only unobserved cells contribute to posterior predictive uncertainty. Applying the law of total variance gives

$$\text{Var} \left(\sum_{j \in \mathcal{E}_i^c} Y_{ij}^* \mid Y_{\mathcal{E}} \right) = \mathbb{E} \left[\text{Var} \left(\sum_{j \in \mathcal{E}_i^c} Y_{ij}^* \mid p, Y_{\mathcal{E}} \right) \mid Y_{\mathcal{E}} \right] + \text{Var} \left[\mathbb{E} \left(\sum_{j \in \mathcal{E}_i^c} Y_{ij}^* \mid p, Y_{\mathcal{E}} \right) \mid Y_{\mathcal{E}} \right].$$

Conditional Bernoulli independence implies

$$\text{Var} \left(\sum_{j \in \mathcal{E}_i^c} Y_{ij}^* \mid p, Y_{\mathcal{E}} \right) = \sum_{j \in \mathcal{E}_i^c} p_{ij}(1 - p_{ij}),$$

and

$$\mathbb{E} \left(\sum_{j \in \mathcal{E}_i^c} Y_{ij}^* \mid p, Y_{\mathcal{E}} \right) = \sum_{j \in \mathcal{E}_i^c} p_{ij}.$$

Dividing by J^2 proves the result. $\square$

Theorem 5.2 (Distributional Error Bound).

$$W_1(\hat{F}, F^*) \leq \text{MAE}_{\text{prompt}}.$$

Proof. Let

$$S_{(1)}^* \leq \dots \leq S_{(I)}^*, \quad \hat{S}_{(1)} \leq \dots \leq \hat{S}_{(I)}$$

denote the realised and recovered prompt scores arranged in increasing order. Therefore,

$$W_1(\hat{F}, F^*) = \int_0^1 |\hat{Q}(u) - Q^*(u)| du = \frac{1}{I} \sum_{i=1}^I |\hat{S}_{(i)} - S_{(i)}^*| \leq \frac{1}{I} \sum_{i=1}^I |\hat{S}_i - S_i^*| = \text{MAE}_{\text{prompt}}$$

$\square$

Proposition 5.2 (Prompt-count balance). *Let*

$$d_i = |\{j \in \mathcal{J} : (i, j) \in \mathcal{E}\}|$$

be the number of observed evaluations for prompt i . Under two-way balanced sampling,

$$|d_i - d_{i'}| \leq 1, \quad i, i' \in \mathcal{I}.$$

Proof. The algorithm always selects a prompt with the smallest current count, so no prompt can exceed another by more than one observation. $\square$

6 Results

6.1 Experimental Setup

We evaluate sparse recovery using MMLU benchmarks. The experiment covers five model splits: Gemma-7B, Llama-3-8B, Llama-3-8B-Instruct, Mistral-7B-v0.1, and Mistral-7B-Instruct-v0.2. For each model, we use ten MMLU subjects and 100 prompt templates. For each budget $B \in \{200, 400, 800, 1600\}$, we generate ten independent masks and compare Avg, PE-Rasch, and BayPIRE on the same observed entries.

6.2 Point and Distributional Recovery

Table 1 gives the aggregate results, averaged over 50 model–subject combinations and ten masks per setting. BayPIRE outperforms all the baseline estimators on prompt MAE, Wasserstein distance, log loss and brier score at every budget. Particularly, BayPIRE reduces prompt MAE from 0.124 for PE-Rasch to 0.027 at $B = 200$. This shows the effectiveness of BayPIRE in the sparse-evaluation recovery.

budget	method	Prompt MAE	W_1	Log loss	Brier
200	Avg	0.284	0.267	4.526	0.343
200	PE-Rasch	0.124	0.104	0.505	0.138
200	BayPIRE	0.027	0.019	0.360	0.105
400	Avg	0.191	0.173	1.869	0.285
400	PE-Rasch	0.086	0.066	0.469	0.113
400	BayPIRE	0.022	0.016	0.266	0.075
800	Avg	0.132	0.114	0.891	0.257
800	PE-Rasch	0.069	0.052	0.375	0.094
800	BayPIRE	0.020	0.014	0.226	0.066
1600	Avg	0.089	0.072	0.692	0.243
1600	PE-Rasch	0.048	0.032	0.265	0.076
1600	BayPIRE	0.019	0.012	0.204	0.061

Table 1 Balanced-mask recovery results averaged over five models, ten MMLU subjects, and ten masks per model–subject–budget setting.

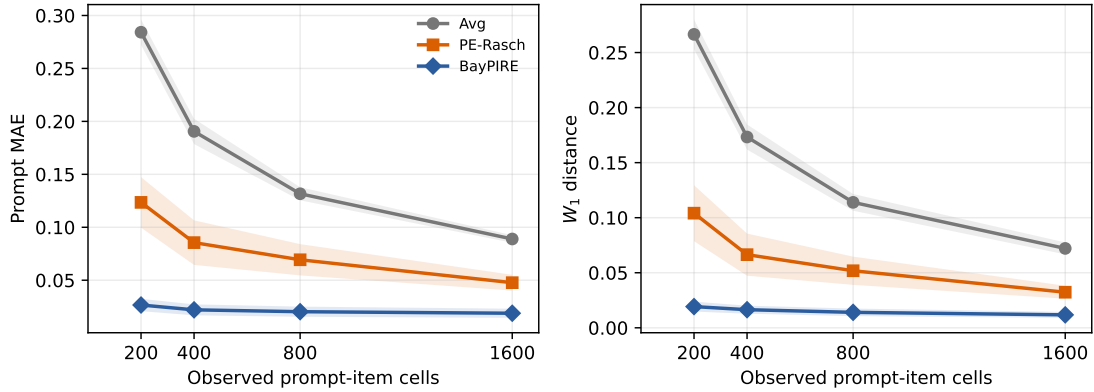


Figure 1 Prompt-level and distributional recovery under two-way balanced masking.

6.3 Hidden-Cell Probability Prediction

BayPIRE also gives the best hidden-cell probability scores. For example, BayPIRE obtains log loss 0.360 and brier score 0.105, compared with 0.505 and 0.138 for PE-Rasch at $B = 200$. This demonstrates the efficacy of BayPIRE over PE-Rasch.

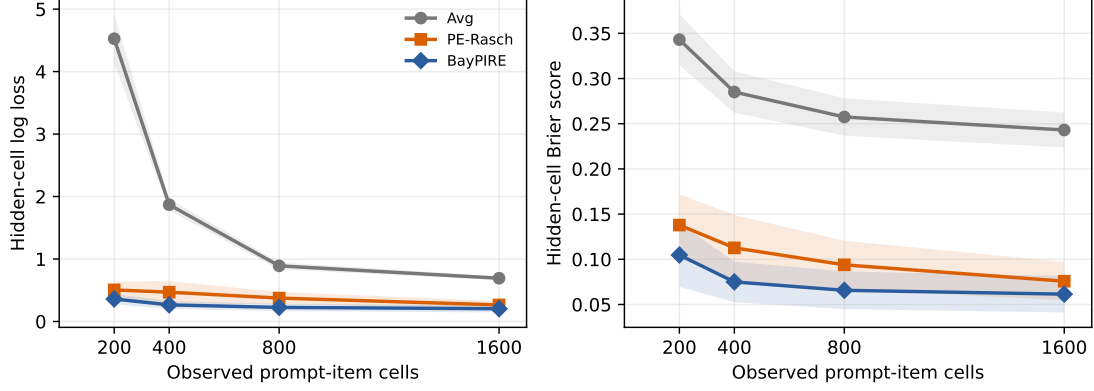


Figure 2 Hidden-cell probabilistic prediction under balanced masking.

6.4 Posterior Predictive Uncertainty

BayPIRE returns posterior predictive intervals for realised prompt scores. Table 2 reports uncertainty summaries under balanced masks. Predictive prompt-score coverage is conservative but stable, decreasing from 1.000 to 0.981 gradually. Mean interval width also decreases monotonically from 0.274 to 0.103, showing more observations do give more information to the estimators.

budget	Prompt coverage	Prompt width	Mean coverage	Mean width	Convergence rate
200	1.000	0.274	0.990	0.092	1.000
400	0.996	0.164	0.974	0.054	1.000
800	0.988	0.125	0.970	0.034	0.990
1600	0.981	0.103	0.960	0.022	0.934

Table 2 BayPIRE posterior predictive uncertainty under balanced masking.

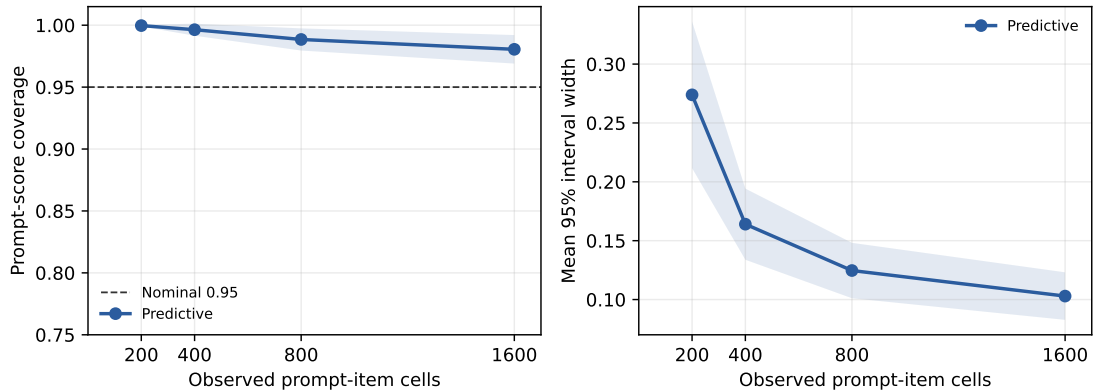


Figure 3 BayPIRE posterior predictive coverage and interval width under balanced masking.

7 Discussion

The experiment supports two conclusions. First, sparse multi-prompt evaluation is a statistical recovery problem. The gap between Avg and the item-response estimators (PE-Rasch and

BayPIRE) shows that item difficulty should be modelled if we only observe a subset of the prompt-item table. Second, full Bayesian posterior averaging improves over the PromptEval Rasch and gives uncertainty intervals whose interpretation can be checked.

BayPIRE’s posterior predictive intervals are conservative, especially at small budgets. However, their width decreases as the budget increases. This is what we desired. This is because when more outcomes are observed, recovered prompt scores should carry less uncertainty and the intervals should become narrower.

7.1 Limitations

The study uses only released PromptEval MMLU results [1] and treats the realised correctness matrix as fixed before masking. This does not model generation-level randomness, answer-extraction uncertainty, or even grader uncertainty. Its posterior predictive intervals are also conservative and may be wider than necessary. This may reduce their practical informativeness. Finally, BayPIRE is computationally more expensive than point-estimation methods because it uses MCMC.

7.2 Future Work

A natural extension is hierarchical BayPIRE. Recent hierarchical Bayesian approaches to LLM evaluation, such as HiBayES [4]. Luettgau et al. [4] suggest that benchmark structure can be used to share statistical strength across related units. In this setting, that could mean partial pooling across prompt families, item categories, MMLU subjects, or model families. Another direction is to include prompt and item covariates. Modelling repeated stochastic decoding is also important: with nonzero temperature, each prompt–item pair may have a distribution of correctness outcomes rather than one realised binary outcome [7].

8 Conclusion

BayPIRE enables more comprehensive multi-prompt evaluation by extending IRT-based recovery from point estimation to uncertainty-aware posterior predictive recovery. In our experiment, BayPIRE improves prompt-score recovery, distributional recovery, and hidden-cell probability prediction over Avg and PE-Rasch. This makes it possible to estimate the distribution of performance with posterior predictive uncertainty across many prompt variants without exhaustive evaluation.

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A Stochastic and Non-Binary Evaluation

The main paper treats $Y_{ij}^* \in \{0, 1\}$ as a realised binary correctness outcome. In stochastic evaluation settings, for example when decoding uses nonzero temperature, the same prompt–item pair may have different outcomes across repeated runs. If each pair is evaluated R times, one may define repeated outcomes Y_{ijr}^* and empirical success rates $\bar{Y}_{ij}^* = R^{-1} \sum_{r=1}^R Y_{ijr}^* \in [0, 1]$.

B Uniform Random Sampling

As a robustness check, we also consider uniform random sampling. Under this design, $\mathcal{E}$ is sampled uniformly without replacement from $\mathcal{I} \times \mathcal{J}$, subject to $|\mathcal{E}| = B$. Unlike two-way balanced sampling, this design does not force equal prompt or item coverage.

Table 4 shows that both balanced and random masks give similar trends. This suggests BayPIRE framework is robust to the sampling design

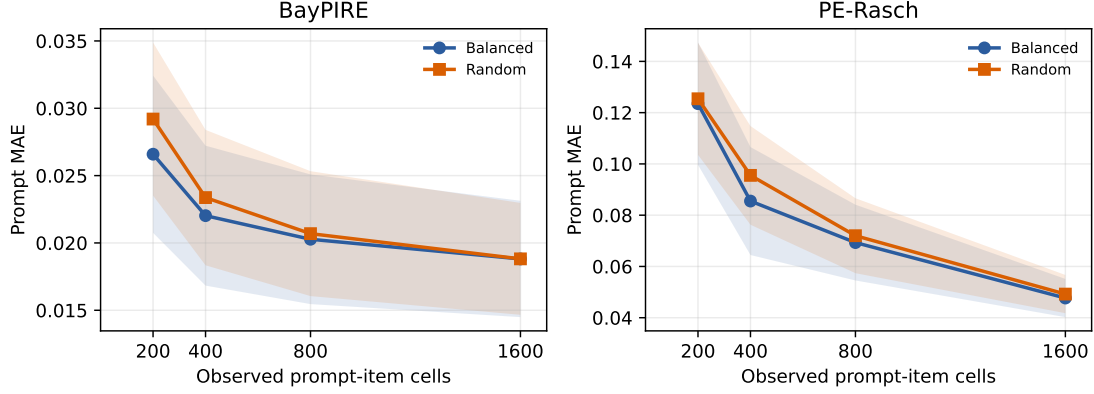


Figure 4 Balanced versus uniform random masking.

budget	method	balanced	random	difference
200	Avg	0.284	0.272	0.012
200	PE-Rasch	0.124	0.125	-0.002
200	BayPIRE	0.027	0.029	-0.003
400	Avg	0.191	0.211	-0.020
400	PE-Rasch	0.086	0.096	-0.010
400	BayPIRE	0.022	0.023	-0.001
800	Avg	0.132	0.138	-0.007
800	PE-Rasch	0.069	0.072	-0.003
800	BayPIRE	0.020	0.021	-0.000
1600	Avg	0.089	0.091	-0.002
1600	PE-Rasch	0.048	0.049	-0.002
1600	BayPIRE	0.019	0.019	-0.000

Table 3 Prompt MAE under balanced and random masks, where difference = balanced minus random.

C Subject-Level Reduction

Figure 5 plots the reduction in prompt MAE from replacing PE-Rasch with BayPIRE under balanced masks. The improvement is positive across all ten subjects and all four budgets.

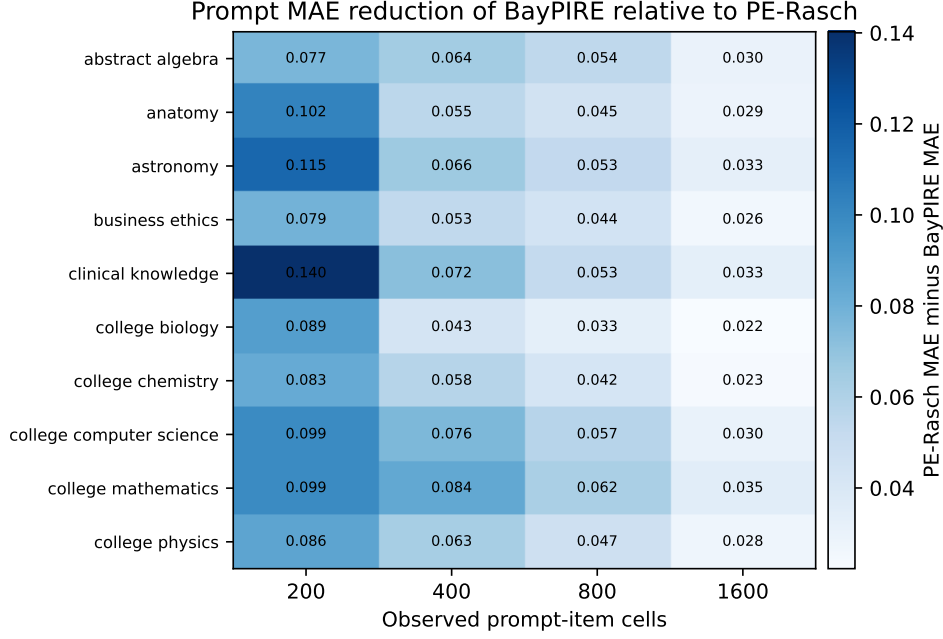


Figure 5 Subject-level prompt MAE reduction of BayPIRE relative to PE-Rasch..

D PE-Rasch Algorithm

Algorithm 2: PE-Rasch point recovery

Input: Observed outcomes $Y_{\mathcal{E}}$, prompt set $\mathcal{I}$, item set $\mathcal{J}$

Output: $\hat{p}_{ij}, \hat{S}_i, \hat{F}, \hat{Q}$

Fit the PE-Rasch model on $Y_{\mathcal{E}}$;

Compute $\hat{p}_{ij}$ for all $(i, j) \in \mathcal{I} \times \mathcal{J}$;

Construct $\hat{Y}$ by observed-retention reconstruction;

Compute $\hat{S}_i = J^{-1} \sum_j \hat{Y}_{ij}$ for all $i \in \mathcal{I}$;

Compute $\hat{F}$ and $\hat{Q}$;

return $\hat{p}, \hat{S}, \hat{F}, \hat{Q}$

E Generative AI Use Statement

During this project, ChatGPT and Codex (OpenAI) was used as an tool for language editing, clarification of mathematical and statistical concepts, assistance with Python and LaTeX code.

Mathematical derivations, methodological choices, experimental design, implementation, analysis of results, and scientific conclusions were checked and determined by the author. No generative AI output was copied directly into the report without independent review and rewriting.

F Reproducibility

All code used to run the experiments, reproduce the reported results, and generate the figures is available at <https://github.com/ish24cho/baypire>. The repository includes the experiment

configurations, sampling settings, random seeds, and scripts used to produce the reported tables and figures.